VHDL code for D Flip Flop

VHDL code for D Flip Flop is presented in this project. Verilog code for D Flip Flop here. There are several types of D Flip Flops such as high-level asynchronous reset D Flip-Flop, low-level asynchronous reset D Flip-Flop, synchronous reset D-Flip-Flop, rising edge D Flip-Flop, falling edge D Flip-Flop, which is implemented in VHDL in this VHDL project.

VHDL code for Rising Edge D Flip Flop:

```
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-- VHDL code for D Flip FLop
-- VHDL code for rising edge D flip flop
Library IEEE;
USE IEEE.Std_logic_1164.all;

entity RisingEdge_DFlipFlop is
   port(
      Q : out std_logic;
      Clk :in std_logic;
      D :in  std_logic
   );
end RisingEdge_DFlipFlop;
architecture Behavioral of RisingEdge_DFlipFlop is
begin
 process(Clk)
 begin
    if(rising_edge(Clk)) then
   Q <= D;
    end if;
 end process;
end Behavioral;
```

VHDL code for Rising Edge D Flip-Flop with Synchronous Reset:

```
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-- VHDL code for D Flip FLop
-- VHDL code for Rising edge D flip flop with Synchronous Reset input
Library IEEE;
USE IEEE.Std_logic_1164.all;

entity RisingEdge_DFlipFlop_SyncReset is
   port(
      Q : out std_logic;
      Clk :in std_logic;
   sync_reset: in std_logic;
      D :in  std_logic
   );
end RisingEdge_DFlipFlop_SyncReset;
architecture Behavioral of RisingEdge_DFlipFlop_SyncReset is
begin
 process(Clk)
 begin
```

```
    if(rising_edge(Clk)) then
   if(sync_reset='1') then
    Q <= '0';
   else
    Q <= D;
   end if;
    end if;
 end process;
end Behavioral;
```

VHDL code for Rising Edge D Flip-Flop with Asynchronous Reset High Level:

```
-- FPGA projects using VHDL/ VHDL
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-- VHDL code for D Flip FLop
-- VHDL code for Rising edge D flip flop with Asynchronous Reset high
Library IEEE;
USE IEEE.Std_logic_1164.all;

entity RisingEdge_DFlipFlop_AsyncResetHigh is
   port(
      Q : out std_logic;
      Clk :in std_logic;
   sync_reset: in std_logic;
      D :in  std_logic
   );
end RisingEdge_DFlipFlop_AsyncResetHigh;
architecture Behavioral of RisingEdge_DFlipFlop_AsyncResetHigh is
begin
 process(Clk,sync_reset)
 begin
     if(sync_reset='1') then
   Q <= '0';
     elsif(rising_edge(Clk)) then
   Q <= D;
  end if;
 end process;
end Behavioral;
```

VHDL code for Rising Edge D Flip-Flop with Asynchronous Reset Low Level:

```
-- FPGA projects using VHDL/ VHDL
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-- VHDL code for D Flip FLop
-- VHDL code for Rising edge D flip flop with Asynchronous Reset low
Library IEEE;
USE IEEE.Std_logic_1164.all;

entity RisingEdge_DFlipFlop_AsyncResetLow is
   port(
      Q : out std_logic;
      Clk :in std_logic;
   sync_reset: in std_logic;
      D :in  std_logic
   );
end RisingEdge_DFlipFlop_AsyncResetLow;
architecture Behavioral of RisingEdge_DFlipFlop_AsyncResetLow is
begin
 process(Clk,sync_reset)
 begin
     if(sync_reset='0') then
   Q <= '0';
     elsif(rising_edge(Clk)) then
   Q <= D;
  end if;
 end process;
end Behavioral;
```

VHDL code for Falling Edge D Flip Flop:

```
-- FPGA projects using VHDL/ VHDL
-- fpga4student.com
-- VHDL code for D Flip FLop
-- VHDL code for falling edge D flip flop
Library IEEE;
USE IEEE.Std_logic_1164.all;

entity FallingEdge_DFlipFlop is
   port(
      Q : out std_logic;
      Clk :in std_logic;
      D :in  std_logic
   );
end FallingEdge_DFlipFlop;
architecture Behavioral of FallingEdge_DFlipFlop is
begin
 process(Clk)
 begin
    if(falling_edge(Clk)) then
   Q <= D;
    end if;
 end process;
end Behavioral;
```

VHDL code for Falling Edge D Flip-Flop with Synchronous Reset:

```
-- FPGA projects using VHDL/ VHDL
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-- VHDL code for D Flip FLop
-- VHDL code for Falling edge D flip flop with Synchronous Reset input
Library IEEE;
USE IEEE.Std_logic_1164.all;

entity FallingEdge_DFlipFlop_SyncReset is
   port(
      Q : out std_logic;
      Clk :in std_logic;
   sync_reset: in std_logic;
      D :in  std_logic
   );
end FallingEdge_DFlipFlop_SyncReset;
architecture Behavioral of FallingEdge_DFlipFlop_SyncReset is
begin
 process(Clk)
 begin
    if(falling_edge(Clk)) then
   if(sync_reset='1') then
    Q <= '0';
   else
    Q <= D;
   end if;
    end if;
 end process;

end Behavioral;
```

VHDL code for Falling Edge D Flip-Flop with Asynchronous Reset High Level:

```
-- FPGA projects using VHDL/ VHDL
-- fpga4student.com
-- VHDL code for D Flip FLop
-- VHDL code for Falling edge D flip flop with Asynchronous Reset high
Library IEEE;
USE IEEE.Std_logic_1164.all;
```

```
entity FallingEdge_DFlipFlop_AsyncResetHigh is
   port(
      Q : out std_logic;
      Clk :in std_logic;
   sync_reset: in std_logic;
      D :in  std_logic
   );
end FallingEdge_DFlipFlop_AsyncResetHigh;
architecture Behavioral of FallingEdge_DFlipFlop_AsyncResetHigh is
begin
 process(Clk,sync_reset)
 begin
     if(sync_reset='1') then
   Q <= '0';
     elsif(falling_edge(Clk)) then
   Q <= D;
  end if;
 end process;
end Behavioral;
```

VHDL code for Falling Edge D Flip-Flop with Asynchronous Reset Low Level:

```
-- FPGA projects using VHDL/ VHDL
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-- VHDL code for D Flip FLop
-- VHDL code for Falling edge D flip flop with Asynchronous Reset low
Library IEEE;
USE IEEE.Std_logic_1164.all;

entity FallingEdge_DFlipFlop_AsyncResetLow is
   port(
      Q : out std_logic;
      Clk :in std_logic;
   sync_reset: in std_logic;
      D :in  std_logic
   );
end FallingEdge_DFlipFlop_AsyncResetLow;
architecture Behavioral of FallingEdge_DFlipFlop_AsyncResetLow is
begin
 process(Clk,sync_reset)
 begin
     if(sync_reset='0') then
   Q <= '0';
     elsif(falling_edge(Clk)) then
   Q <= D;
  end if;
 end process;
end Behavioral;
```
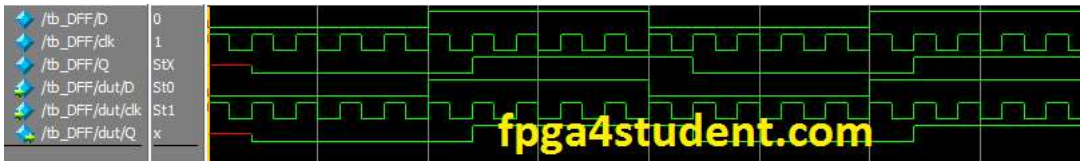
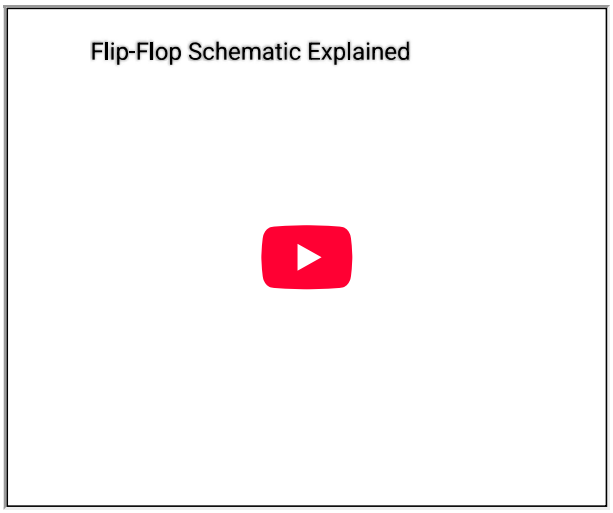

Simulation waveform for D Flip-Flop:



This tutorial explains in detail the flip-flop schematic and how it works: HERE.



To understand more about setup and hold time of flip-flop, check it out HERE.
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